

Interpretation of Statistical Graphs

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Statistical graphs are visual representations of data that make information easier to understand.

Bar Graph

A **bar graph** compares quantities among different categories.

Ask: Which bar is tallest? Which is shortest?

Line Graph

A **line graph** shows how data changes over time.

Ask: Is the line going up, going down, or staying flat?

Pie Chart

A **pie chart** shows how a whole is divided into parts or percentages.

Ask: Which slice is the biggest? Which is the smallest?

Histogram

A **histogram** displays the distribution of grouped numerical data.

Ask: Which interval (range) has the tallest bar?

What Does “Interpret” Mean?

To **interpret** a graph means to:

- ▶ Read the information
- ▶ Analyze it
- ▶ Compare values
- ▶ Draw conclusions

What to Look For

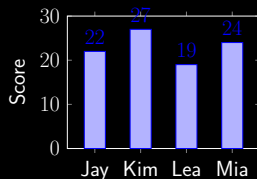
When interpreting a graph, look for the:

- ▶ **Greatest** value
- ▶ **Least** value
- ▶ **Trends** (increasing/decreasing)
- ▶ **Differences** between values
- ▶ Possible **conclusions** supported by the data

Examples

Example 1

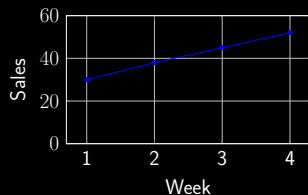
Which student scored the highest?



Kim has the tallest bar, so Kim scored the **highest**.

Example 2

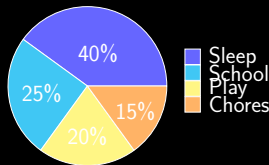
Describe the trend in weekly sales.



The line keeps going up, so sales show an **increasing trend**.

Example 3

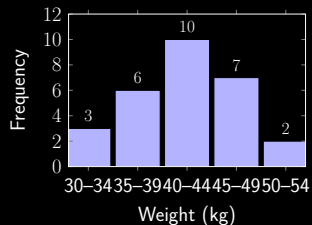
Which activity takes up the largest portion of the day?



The largest slice is **Sleep** at 40%.

Example 4

Which weight interval has the highest frequency?

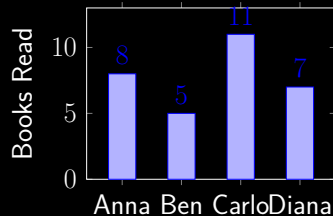


The interval 40–44 has the tallest bar, so it has the **highest frequency**.

Practice Exercises

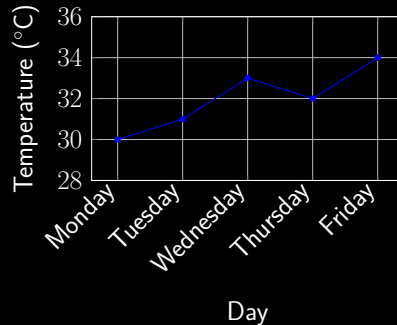
Practice 1

A bar graph shows the number of books read by four students. Who read the most books?



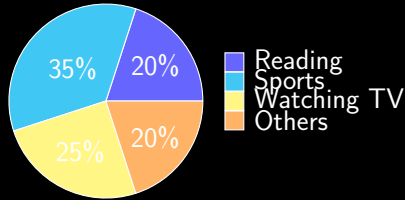
Practice 2

Refer to the line graph of daily temperatures.
On which day was the highest temperature recorded?



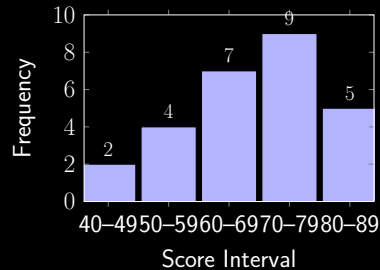
Practice 3

A pie chart shows how a student spends free time. Which activity occupies the largest portion?



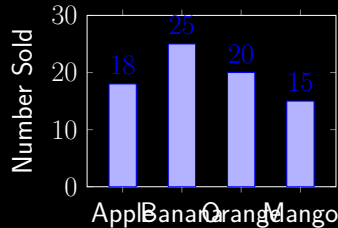
Practice 4

A histogram shows the following grouped scores. Which score interval has the highest frequency?



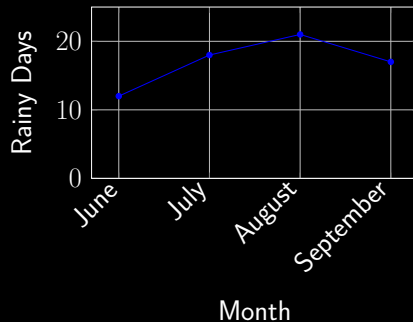
Practice 5

A bar graph shows the number of fruits sold. How many more bananas were sold than mangoes?



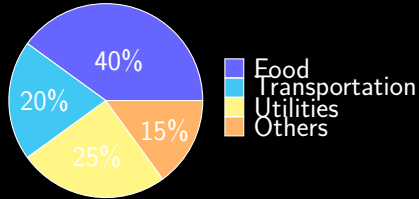
Practice 6

Refer to the line graph. During which month did the number of rainy days decrease?



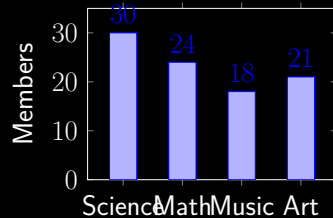
Practice 7

A pie chart shows the monthly expenses of a family. Which expense takes the smallest percentage?



Practice 8

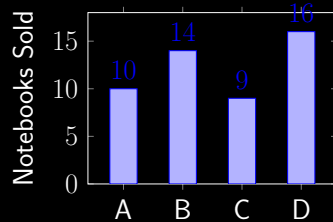
Study the following bar graph data. Which club has the second largest number of members?



Activities

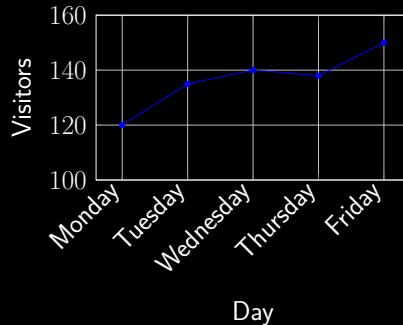
Activity 1

A bar graph shows the number of notebooks sold. Which store sold the most notebooks?



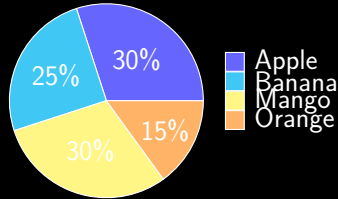
Activity 2

Refer to the line graph of daily visitors. On which day was the greatest number of visitors recorded?



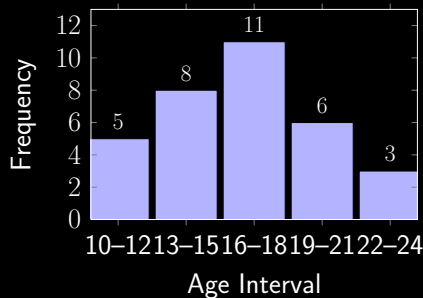
Activity 3

A pie chart shows favorite fruits. Which fruit has the smallest percentage?



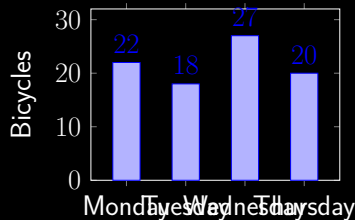
Activity 4

A histogram shows the following grouped ages. Which age interval has the highest frequency?



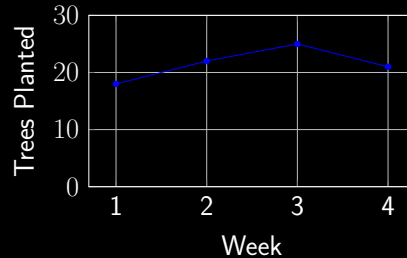
Activity 5

A bar graph shows the number of bicycles rented. How many more bicycles were rented on Wednesday than Tuesday?



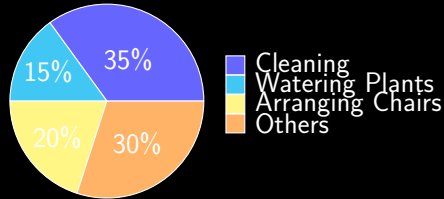
Activity 6

Refer to the following line graph. During which week did the number of trees planted decrease?



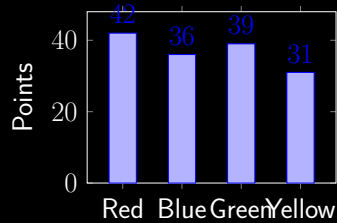
Activity 7

A pie chart shows classroom responsibilities.
Which task has the smallest percentage?



Activity 8

Study the following bar graph data. Which team has the second highest number of points?



Thank you!